

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and applied quantitative researcher specializing in stochastic processes, statistical modeling of sparse and noisy dependent data, numerical optimization, and simulation. Develops Python/PyTorch prototypes that connect mathematical models with empirical diagnostics, scalable computation, and risk decisions.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Dec 2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2019 - Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Computing: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), R, SQL/MySQL, Linux, Git, AWS EC2, HPC/SLURM

Mathematical and Statistical Methods: Probability, stochastic processes, Gaussian processes, MLE/GLS, numerical optimization (L-BFGS), Monte Carlo simulation, spectral methods, predictive modeling, missing-data and model diagnostics

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling

Summer 2026

- Developed a sequential anomaly-detection framework for binary risk indicators by representing cumulative evidence as a random walk and using Brownian-motion approximations to quantify stopping-time uncertainty.
- Used simulation to compare fixed-sample and sequential policies across detection delay, false-positive/false-negative control, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM predictive-modeling pipeline for mid-market property risk, including data processing, GLM benchmarking, validation, and diagnostics across 2.46M+ records on AWS EC2.
- Explained model diagnostics and feature-attribution results to business stakeholders and translated them into pricing and risk-segmentation recommendations.

APPLIED RESEARCH AND TRADING PROJECTS

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and evaluated a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven performance analysis.

Ph.D. Dissertation Research - Scalable Gaussian Process Inference and Diagnostics

Sep 2024 - Present

- Sparse and Noisy Data Modeling:** Developed an advection-aware Vecchia approximation for nonseparable Gaussian processes with structured missingness, replacing dense $O(n^3)$ covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- Estimation and Optimization:** Calibrated Matérn and generalized Cauchy covariance models using GLS mean profiling and L-BFGS; designed block-prefix experiments to separate approximation error from covariance-family misspecification.
- Simulation and Assumption Testing:** Validated a mask-aware expected cross-periodogram over 1,000 Monte Carlo simulations (MAD 0.00648) and built scale- and frequency-resolved diagnostics for missing-data and covariance misspecification.
- Scalable Prototyping:** Implemented reusable Python/PyTorch likelihood and diagnostic pipelines with CPU/GPU execution, cached fit artifacts, and restartable HPC jobs.

ADDITIONAL

Coursework: Machine Learning, Linear Algebra, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics, Econometrics

Languages: Korean (Native); English (Fluent); Chinese (Basic)